CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

3rd Semester of B. Pharm Examination

University Theory Examination November – December 2017

PH203 Organic Chemistry-I

Date: 24/11/2017, Friday Time: 10:00 a.m to 01:00 p.m Maximum Marks: 80

Instructions:

1. **Figures to right indicate marks.**
2. **Section wise separate answer book is to be used.**
3. **Draw neat sketches wherever necessary.**

SECTION- I

Q 1		Attempt any **TWO** of the following:	
A	1.	Discuss different types of elimination reactions with the help of suitable examples.	**05**
	2.	Write a note on Hyperconjugation.	**05**
B	1.	What is tautomerism? Explain with suitable examples.	**05**
	2.	Write a note on Hoffman's orientation.	**05**
C	1.	Write a note on Halogenation of alkane.	**05**
	2.	Explain elimination versus substitution.	**05**
Q 2		Attempt any **TWO** of the following:	
A	1.	Discuss in detail the intermolecular forces operating in organic compounds.	**05**
	2.	Write a note on saytzaff's rule.	**05**
B	1.	What is hybridization? Explain SP^2 hybridization.	**05**
	2.	Write a note Diel's Alder reactions along with it's mechanism.	**05**
C	1.	Write a note on Inductive effect.	**05**
	2.	Explain Kjeldahl's Method in details.	**05**

SECTION- II

Q 3 Attempt any **TWO** of the following:

A **Define following terms with examples (Any Five) (05 X 02 = 10)** **10**

i. Racemic Mixture

ii. Intermolecular Forces

iii. Meso Compounds

iv. E and Z configuration

v. Optical Activity

vi. Cis and Trans

B Explain reaction, mechanism, reactivity, kinetic and stereochemistry of SN_1 and SN_2 reaction. **10**

C 1. Explain the preparation of ether and epoxide. **05**

2. Give any three reactions of alkene and alkynes. **05**

Q 4 Attempt any **TWO** of the following:

A What do mean by rate determining step? Discuss in detail about mechanism, reactivity, kinetic and orientation of E_1 and E_2 reactions. **10**

B 1. What is conformation? Explain the conformation of n-Butane with energy diagram. **05**

2. Explain the preparation and reaction of alcohol. **05**

C Discuss the various factor affecting SN_1 and SN_2 reaction. **10**
